Nishchal Thakur

B.Tech Electronics & Communication Engineering Student | Software Engineering & DSA

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PROFESSIONAL SUMMARY

B.Tech Electronics & Communication Engineering student with hands-on experience in C++, Data Structures & Algorithms, graph algorithms, React.js, JavaScript, and SQL. Built a C++ pathfinding engine using Dijkstra's algorithm and a React-based analytics dashboard for competitive programming. Strong foundation in object-oriented programming, operating systems, and algorithmic problem solving.

EDUCATION

University Institute of Technology, RGPV — Bhopal, Madhya Pradesh

B.Tech in Electronics & Communication Engineering — 2023 – 2027 | CGPA: 7.85/10

PROJECTS

Smart City Pathfinding Engine — GitHub

Tech: C++, Object-Oriented Programming, Graph Theory

- Designed a modular object-oriented system to model city intersections and road networks using encapsulated C++ classes.
- Implemented a custom min-heap from scratch with dynamic arrays and parent/child index calculations, supporting **O(log V)** extraction operations.
- Built an undirected, weighted graph with adjacency-list representation and integrated Dijkstra's algorithm with the custom heap to compute optimal routes.

CodeTrack — DSA & Competitive Programming Analytics Dashboard — GitHub

Tech: React.js, JavaScript, HTML5, CSS3, LocalStorage

- Developed a responsive dashboard to track coding progress, solved problems, streaks, platform-wise statistics, and DSA topic completion.
- Implemented search, filtering, sorting, difficulty/topic tracking, custom revision lists, and persistent LocalStorage data management.
- Built activity calendars, streak calculations, monthly progress analytics, and reusable React components for competitive programming workflows.

TECHNICAL SKILLS

Languages: C++, JavaScript, SQL

Frontend: HTML5, CSS3, React.js

CS Fundamentals: Data Structures & Algorithms, OOP, Operating Systems

Tools: Git, GitHub, VS Code, Vite

ACHIEVEMENTS

- Solved **400+ problems** on LeetCode; current rating: **1600**.
- Built a strong foundation in Data Structures & Algorithms using C++.
- Participated in coding contests and regularly practice algorithmic problem solving.